Shaun John Thomas

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PROFILE

Final-year Aero-Mechanical Engineer with strong aerodynamic fundamentals and hands-on experience in Computational Fluid Dynamics (CFD) and stability analysis, currently applying this to the design and upcoming manufacture and flight testing of a Blended Wing Body Unmanned Aerial Vehicle (UAV). Directed a 4-member team through the design, Geometric Dimensioning and Tolerancing (GD&T)-based manufacturing and build of a 10,000 ft model rocket, coordinating sub-systems from structures to propulsion. Proficient across the full simulation-to-hardware pipeline, with proven cross-functional collaboration in multidisciplinary student engineering teams.

EDUCATION

University

BEng Aero-Mechanical Engineering, University of Strathclyde — Glasgow, UK | September 2025- Present
Current Average: 69%

BTech Mechanical Engineering, Vellore Institute of Technology (Transferred) — Chennai, India | August 2023 – June 2025
2 Year CGPA: 8.4/10

EXPERIENCE AND PROJECTS

Blended Wing Body UAV Project — Glasgow, UK | August 2026 - Present
Individual Project

- Carrying out a parametric study quantifying the effect of winglet cant angle on Dutch roll damping in a rudderless Blended Wing Body UAV, benchmarking results against MIL-F-8785C and CS-23 qualities criteria.
- Extracting stability derivatives via vortex-lattice modelling (XFLR5, AVL) and building a Python state-space eigenvalue model to predict lateral-directional dynamics across wingtip configurations.
- Validating methodology and derivative outputs against published VLM stability studies on tailless configurations.
- Preparing to manufacture a 3D printed and composite reinforced airframe and winglets with varying cant angles for flight-test validation.

Airframe Design Engineer — Glasgow, UK | September 2025 - Present
StrathAIS, Student Rocketry Team, University of Strathclyde

- Engineering and manufacturing bulkhead and supporting structures for a 30,000 ft, Mach 1+ rocket, presenting design and verification evidence through PDR and CDR reviews for IREC 2026.
- Running FEA simulations to shape critical design choice and maximize strength to weight ratio.
- Recommending and implementing changes in manufacturing decisions to minimize error and material waste with the experience gained from previous failures.

Airframe Lead — Chennai, India | August 2023 – June 2025
Team Ignition, Student Rocketry Team, Vellore Institute of Technology

- Directed a team of 4 to design and build the airframe of a model rocket to reach an altitude of 10,000 feet, collaborating closely with various departments to reach innovative design compromises across the project.
- Manufactured supporting structures using power tools and CNC machines and reinforced rocket fins and nose cones with fiberglass sheets to withstand high temperatures and harsh conditions.
- Validated design choices using a testing suite, including CFD, FEA, pressure and vibration analysis with the help of technical guidance from external manufacturers on industry standards and practices.

SKILLS & CERTIFICATIONS

Skills and Soft Skills:

- CAD: SolidWorks, OnShape
- Manufacturing: 3D-Printing, Composites, GD&T
- Technical leadership & cross-department collaboration
- Project Management & Problem Solving
- Simulations: ANSYS (CFD & FEA), XFLR5, OpenFOAM
- Machining: CNC machining, power tools
- Numerical Analysis: MATLAB & SIMULINK, Python
- Design Reviews & Technical Communication

Certifications: Certified SolidWorks Design Associate (CSWA)